

Expert Behavioral Categories for SegPro-Med

Overview

This document defines meaningful behavioral categories derived from user interaction patterns in the medical imaging annotation platform. Categories are evidence-based and provide actionable insights for workflow optimization, training, and quality control.

1. ANNOTATION APPROACH CATEGORIES

1.1 AI Dependency Level

Purpose: Understand reliance on AI-assisted segmentation

- **AI-Heavy User** (AI Assistance Rate > 60%)
 - Frequently uses MedSAM2, MedGemma, Med-R1, SmolVLM
 - High acceptance rate of AI suggestions (>70%)
 - Lower manual annotation time per slice
 - *Insight:* Efficient but may need quality validation checks
- **Hybrid User** (AI Assistance Rate 30-60%)
 - Balanced use of manual + AI tools
 - Selective AI model usage based on task
 - Moderate AI acceptance rate (40-70%)
 - *Insight:* Strategic, context-aware approach
- **Manual-First User** (AI Assistance Rate < 30%)
 - Prefers manual annotation tools
 - Low AI usage or primarily uses AI for suggestions only
 - High manual annotation time
 - *Insight:* High precision but potentially slower workflow

1.2 Tool Preference Profile

Purpose: Identify preferred annotation drawing methods

- **Rectangle Specialist**
 - Primary tool: `drawrect` (>60% usage)
 - Fast bounding box annotations
 - Suitable for structured ROI marking
- **Freehand Annotator**
 - Primary tools: `drawclosedpath`, `drawopenpath` (>50% combined)
 - Irregular shape segmentation
 - Higher precision for complex anatomical structures
- **Multi-Tool Expert**
 - Uses 4+ different tools regularly
 - Tool selection varies by anatomy/pathology

- Adapts approach to task requirements
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2. EXPERTISE & EFFICIENCY CATEGORIES

2.1 Speed Profile

Purpose: Measure annotation efficiency

- **Speed Demon** (< 45 sec/annotation avg)
 - Rapid decision-making
 - Minimal exploration time
 - High tool proficiency
 - *Risk:* May sacrifice accuracy for speed
- **Balanced Annotator** (45-120 sec/annotation)
 - Moderate pace with quality focus
 - Standard exploration patterns
 - Balanced speed/accuracy tradeoff
- **Meticulous Reviewer** (> 120 sec/annotation)
 - Extended exploration and review
 - Multiple edits per annotation
 - High quality but lower throughput
 - *Opportunity:* Potential training candidate for faster workflows

2.2 Experience Level

Purpose: Identify proficiency patterns

- **Novice**
 - High slice-to-slice variation
 - Frequent tool changes mid-task
 - Extended exploration phases
 - High edit-to-create ratio (>0.4)
 - **Intermediate**
 - Consistent workflow patterns
 - Moderate tool switching
 - Balanced exploration/annotation phases
 - Moderate edit rate (0.2-0.4)
 - **Expert**
 - Highly consistent workflow
 - Minimal tool switching
 - Efficient exploration → annotation transition
 - Low edit rate (<0.2)
 - Rapid label assignment
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3. DOMAIN SPECIALIZATION CATEGORIES

3.1 Modality Expertise

Purpose: Identify imaging modality specialization

- **Brain MRI Specialist**
 - Primary dataset: **brain** (>70% of work)
 - Modalities: **flair, t1, t1c, t2**
 - Labels: Tumor, Edema, Necrosis, Ventricles
 - Views: Heavy axial usage with coronal cross-referencing
- **Mammography Expert**
 - Primary dataset: **MG** (>60% of work)
 - Modality: **MG** (Mammography)
 - Views: LCC, RCC, LML0, RML0
 - Labels: Mass, Calcification, Architectural Distortion
- **Abdominal Imaging Specialist**
 - Primary dataset: **abdomen** (>60% of work)
 - Multi-organ annotation
 - Liver, kidney, spleen focus
 - Labels: Organ, Lesion, Tumor
- **Generalist**
 - Works across 3+ dataset types
 - Adaptable to different modalities
 - Broad anatomical knowledge

3.2 Pathology Focus

Purpose: Understand annotation content expertise

- **Oncology Focused**
 - Primary labels: Tumor, Necrosis, Edema (>60% annotations)
 - Tumor measurement and characterization
 - Multi-slice tracking of lesions
 - **Structural/Anatomical Focused**
 - Primary labels: Organ, ROI, Normal Tissue (>60%)
 - Organ delineation emphasis
 - Anatomical landmark identification
 - **Mixed Pathology Annotator**
 - Diverse label usage (>5 unique labels)
 - Balanced pathology + anatomy
 - Comprehensive case assessment
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4. WORKFLOW PATTERN CATEGORIES

4.1 Navigation Style

Purpose: Understand exploration behavior

- **Sequential Navigator**
 - Linear slice progression (>70% sequential)
 - Minimal slice jumping
 - Systematic coverage
 - *Insight:* Thorough but may miss scattered findings
- **Jump Scanner**
 - Frequent non-sequential navigation (>40%)
 - Targets specific regions
 - Efficient for known pathology
 - *Risk:* May miss subtle findings
- **View Switcher**
 - Frequent view orientation changes (>10 switches/session)
 - Multi-planar review (axial sagittal coronal)
 - Comprehensive spatial understanding
 - *Insight:* High quality but time-intensive

4.2 Review Behavior

Purpose: Assess quality control practices

- **Edit-Heavy Reviewer**
 - High annotation edit rate (>30% of annotations edited)
 - Iterative refinement approach
 - Quality-focused workflow
 - **One-Pass Annotator**
 - Low edit rate (<10%)
 - Confident initial annotations
 - Minimal revision
 - *Risk:* May miss errors without review
 - **Deletion-Prone User**
 - High deletion rate (>15% annotations deleted)
 - Experimental approach or uncertainty
 - May need additional training or clearer guidelines
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5. INTERACTION PATTERN CATEGORIES

5.1 Window/Level Adjustment Behavior

Purpose: Understand image optimization habits

- **Window Tweaker**

- Frequent window/level adjustments (>5 per slice)
- Fine-tunes visualization
- Optimal contrast seeking
- *Insight:* Detail-oriented, may benefit from presets
- **Default Settings User**
 - Minimal window adjustments (<2 per slice)
 - Relies on default visualization
 - *Opportunity:* Training on window optimization

5.2 VLM Usage Pattern

Purpose: Track Vision-Language Model interaction

- **VLM Power User**
 - Regular use of MedGemma/Med-R1/SmolVLM (>20% sessions)
 - Custom prompt creation
 - Voice-to-text prompt generation
 - Leverages label suggestions
 - **VLM Occasional User**
 - Sporadic VLM usage (5-20% sessions)
 - Uses predefined prompts
 - Selective application
 - **VLM Non-User**
 - No VLM engagement
 - Traditional annotation approach only
 - *Opportunity:* VLM training and adoption
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6. COLLABORATION & QUALITY CATEGORIES

6.1 Crowdsourcing Participation

Purpose: Assess engagement in collaborative annotation

- **High-Volume Contributor**
 - 10 assignments completed
 - Consistent submission rate
 - High completion percentage (>80%)
- **Selective Participant**
 - Moderate assignment completion (40-80%)
 - Variable submission timing
 - May have capacity constraints
- **Assignment Struggler**
 - Low completion rate (<40%)
 - Extended time per assignment
 - May need support or workflow adjustment

6.2 Consistency Score

Purpose: Measure annotation quality and reliability

- **Highly Consistent** (Consistency Score >0.85)
 - Reproducible annotation sizes/positions
 - Stable label usage patterns
 - Minimal inter-session variation
 - **Moderately Consistent** (0.7-0.85)
 - Some variation in approach
 - Generally reliable patterns
 - **Inconsistent** (<0.7)
 - High variability in annotations
 - Unstable workflow patterns
 - *Action:* Targeted training or guideline review
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7. EFFICIENCY METRICS CATEGORIES

7.1 Session Productivity

Purpose: Overall workflow efficiency

- **High Efficiency** (Efficiency Score >0.8)
 - High annotations per minute
 - Low idle time
 - Streamlined workflows
 - Minimal errors/deletions
 - **Medium Efficiency** (0.6-0.8)
 - Balanced pace
 - Moderate workflow optimization
 - Room for improvement
 - **Low Efficiency** (<0.6)
 - Extended idle periods
 - Frequent workflow interruptions
 - High error correction time
 - *Action:* Workflow analysis and optimization
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CATEGORY COMBINATIONS FOR PERSONA GENERATION

Example Expert Profiles:

Persona 1: “The AI-Powered Brain Specialist” - AI-Heavy User + Brain MRI Specialist + Speed Demon + VLM Power User - *Characteristics:* Leverages AI extensively for rapid brain tumor annotation, highly productive

Persona 2: “The Meticulous Mammography Expert” - Manual-First User + Mammography Expert + Meticulous Reviewer + Freehand Annotator - *Characteristics:* Precise manual annotation of breast lesions, quality over speed

Persona 3: “The Generalist Hybrid” - Hybrid User + Generalist + Balanced Annotator + Multi-Tool Expert + View Switcher - *Characteristics:* Versatile across modalities, strategic AI use, thorough multi-planar review

Persona 4: “The Novice Learning Curve” - Manual-First User + Novice + Edit-Heavy Reviewer + Sequential Navigator - *Characteristics:* Learning phase, improving through iteration, systematic approach

ACTIONABLE INSIGHTS FROM CATEGORIES

For Training:

- Identify novices for onboarding programs
- Match experts with specializations for mentorship
- Target VLM non-users for AI adoption training

For Workflow Optimization:

- Provide tool presets for Rectangle Specialists
- Create multi-view templates for View Switchers
- Optimize AI pipelines for AI-Heavy Users

For Quality Control:

- Flag Inconsistent annotators for review
- Validate Speed Demons’ work more frequently
- Leverage Highly Consistent users as gold standard

For Resource Allocation:

- Assign brain cases to Brain MRI Specialists
- Match high-volume contributors with urgent campaigns
- Provide additional support to Assignment Strugglers

Last Updated: November 10, 2025 **Version:** 1.0 **Data Source:** SegPro-Med User Behavior Analytics